

Multivariate Analysis of Energy-Consumption and Power-Quality Behaviour in the Steel Industry

Prepared by :Zainab Rashad

EXECUTIVE SUMMARY

This report presents a comprehensive multivariate and predictive analysis of steel industry energy consumption and power quality data. The dataset contains continuous measurements of energy usage, carbon emissions, reactive power, power factor behaviour, operational duration, and temporal and operational indicators. Five analytical questions were investigated using Principal Components Analysis (PCA), clustering methods (hierarchical and K-means), Linear and Quadratic Discriminant Analysis (LDA/QDA), Canonical Correlation Analysis (CCA), and stepwise regression modelling (GLMSELECT) for forecasting.

PCA revealed that the dominant sources of variation in the dataset arise from two major dimensions: overall energy consumption intensity and electrical power quality behaviour, jointly explaining 85.44% of total variability. Clustering identified three distinct operational groups of steel plants, representing high load, moderate load, and low load energy usage profiles. LDA and QDA demonstrated that these groups can be classified with very high accuracy, confirming that the clusters represent statistically meaningful and well separated operational categories.

CCA further revealed a strong multivariate association between energy consumption variables and power quality variables, indicating that plants with higher energy usage also exhibit distinct reactive power and power factor patterns. In addition, the GLMSELECT regression model demonstrated that electricity consumption (usage-kwh) can be predicted with very high accuracy ($R^2 = 0.9804$) using operational, electrical, environmental, and temporal variables. This confirms that energy demand in steel manufacturing is highly predictable and strongly dependent on load conditions and power quality behaviour.

Overall, these findings provide a detailed and integrated understanding of energy behaviour in steel operations, highlighting not only the interdependence between consumption and electrical efficiency but also the strong potential for accurate energy demand forecasting to support operational planning and cost optimization.

Together, these methods provide a unified multivariate understanding of operational behaviour, electrical efficiency, and energy demand.

INTRODUCTION AND RESEARCH QUESTIONS

Energy consumption and electrical power quality performance are central operational indicators in the steel industry. High energy usage may reflect increased production, but it may also indicate inefficiencies in electrical systems, such as excessive reactive power or poor power factor behaviour. Understanding how these variables interact requires a multivariate approach capable of capturing the underlying structure of the data and revealing patterns that are not visible through univariate analysis.

The dataset used in this study contains continuous measurements of energy usage (kWh), carbon emissions, reactive power (lagging and leading), power factor behaviour, and non solar minutes. These variables collectively describe both the magnitude of energy consumption and the quality of electrical power usage. The analysis addresses four research questions:

Q1- What underlying patterns and major sources of variation exist in steel industry's energy consumption data?

Q2- Do steel plants form distinct groups based on their energy usage behaviour, and how do these groups differ in their operational characteristics?

Q3- How accurately can these groups be identified in new observations, and how well can a classification model distinguish between them?

Q4- How are energy consumption variables related to power quality variables and what insights can be drawn about their interdependence?

Q5: How can future electricity consumption (usage kwh) be predicted to support energy demand planning and cost optimization in the steel industry?

2- VARIABLES AND MEASUREMENT SCALE

2.1 Variables and Measurement Scale

The steel industry dataset contains 35,040 observations and seven continuous operational variables, all measured on ratio scales. These variables capture both the magnitude of energy consumption and the behaviour of electrical power quality within industrial operations. The primary energy related variables include *usage-kwh*, which measures total electrical energy consumed; *co2-tco2*, which quantifies carbon emissions associated with energy use; and *nsm*, which records non solar minutes and reflects the duration of plant operation. The remaining variables describe electrical power quality behaviour. Lagging-current-reactive-power-k and leading-current-reactive-power-k measure reactive power under inductive and capacitive load conditions, respectively, while *lagging-current-power-factor* and *leading-current-power-factor* quantify the efficiency with which electrical power is converted into useful work. Because these variables are measured in heterogeneous units kilowatt-hours, tonnes of CO₂, kilovolt-ampere reactive power, and percentage power-factor values standardisation was required before applying PCA, clustering, and discriminant analysis. Descriptive statistics were first computed on the raw variables to understand their natural scales, after which all variables were standardised to mean zero and unit variance to ensure comparability across multivariate methods.

Table 1. Descriptive statistics before scaling (N = 35,040)

Variable	Mean	Std Dev	Minimum	Maximum
usage_kwh	27.3869	33.4444	0	157.18
lagging_current_reactive_power_k	13.0354	16.3060	0	96.91
leading_current_reactive_power_k	3.8709	7.4245	0	27.76
co2_tco2	0.0115	0.0162	0	0.07
lagging_current_power_factor	80.5781	18.9213	0	100
leading_current_power_factor	84.3679	30.4565	0	100
nsm	42750.00	24940.53	0	85500

2.2 Data Preparation Decisions

Given the heterogeneous measurement scales of the variables, standardisation was essential prior to PCA, clustering, and discriminant analysis. All numeric variables were transformed to have a mean of zero and a standard deviation of one, ensuring that no single variable dominated the multivariate structure due to its scale. The dataset did not include a date or timestamp variable, meaning that temporal forecasting could not be performed. Categorical variables such as *day of week*, *weekstatus*, and *load type* were excluded from PCA and clustering because these methods require continuous numeric inputs. The analysis therefore focused exclusively on the seven standardised numeric variables.

Table 2. Descriptive statistics after scaling (standardised variables)

Variable	Mean	Std Dev	Minimum	Maximum
usage_kwh	0.0000	1.0000	−0.8189	3.8809
lagging_current_reactive_power_k	0.0000	1.0000	−0.7994	5.1438
leading_current_reactive_power_k	0.0000	1.0000	−0.5214	3.2176
co2_tco2	0.0000	1.0000	−0.7135	3.6206
lagging_current_power_factor	0.0000	1.0000	−4.2586	1.0265
leading_current_power_factor	0.0000	1.0000	−2.7701	0.5133
nsm	0.0000	1.0000	−1.7141	1.7141

3. METHODS

3.1 Principal Components Analysis (PCA)

Principal Components Analysis was applied to the seven standardised numeric variables to identify the dominant dimensions of variation in the steel industry dataset. Because the variables were measured in heterogeneous units, PCA was performed on the correlation matrix, ensuring that each variable contributed equally to the analysis. The procedure extracted seven components, with eigenvalues and scree plots used to determine the number of components to retain. The first two components together explained 85.44% of the total variance, indicating a highly structured dataset. Component loadings were examined to interpret the underlying operational dimensions represented by each component. PCA scores from the first two components were saved and later used as inputs for clustering to reduce multicollinearity and improve cluster stability.

3.2 K-Means Clustering

Clustering was performed using the FASTCLUS procedure on the first two principal component scores. Using PCA scores rather than raw variables ensured that clustering was based on the most meaningful dimensions of variation and avoided distortions caused by correlated inputs. Solutions for $K = 2, 3, 4$, and 5 were evaluated. The optimal number of clusters was selected using the Pseudo-F statistic, overall R-square, and interpretability of cluster profiles. The $K = 3$ solution emerged as the most statistically and operationally meaningful model, producing well separated clusters with stable sizes and clear differences in energy consumption and power quality behaviour. Cluster means were computed on the original variables to provide operational interpretation of each group.

3.3 Linear and Quadratic Discriminant Analysis (LDA/QDA)

Discriminant analysis was used to evaluate how accurately the identified clusters could be classified for new observations. LDA assumes equal covariance matrices across groups, while QDA allows each group to have

its own covariance structure. Both models were fitted using the seven standardised numeric variables as predictors and the $K = 3$ cluster assignments as the grouping variable. Classification accuracy was assessed using same sample resubstitution tables. Although resubstitution accuracy tends to overestimate true predictive performance, it provides an upper bound on classification capability. Box's M test was used to assess the equality of covariance matrices, guiding the interpretation of LDA versus QDA performance. Posterior probabilities were examined to identify borderline observations and assess model confidence.

3.4 Canonical Correlation Analysis (CCA)

Canonical Correlation Analysis was conducted to examine the multivariate relationship between energy consumption variables and power quality variables. The energy block consisted of usage-kwh, co2-tco2, and nsm, while the power quality block included reactive-power and power-factor variables. CCA extracted canonical functions that maximised the correlation between linear combinations of the two variable sets. The significance of each canonical function was evaluated using Wilks' Lambda and F-tests. Structure loadings were used to interpret the contribution of each variable to its canonical variate. Redundancy analysis quantified how much variance in one block could be explained by the canonical variates of the other block. This allowed assessment not only of correlation strength but also of practical explanatory power.

4-METHODS

4.1 Principal Components Analysis (Q1)

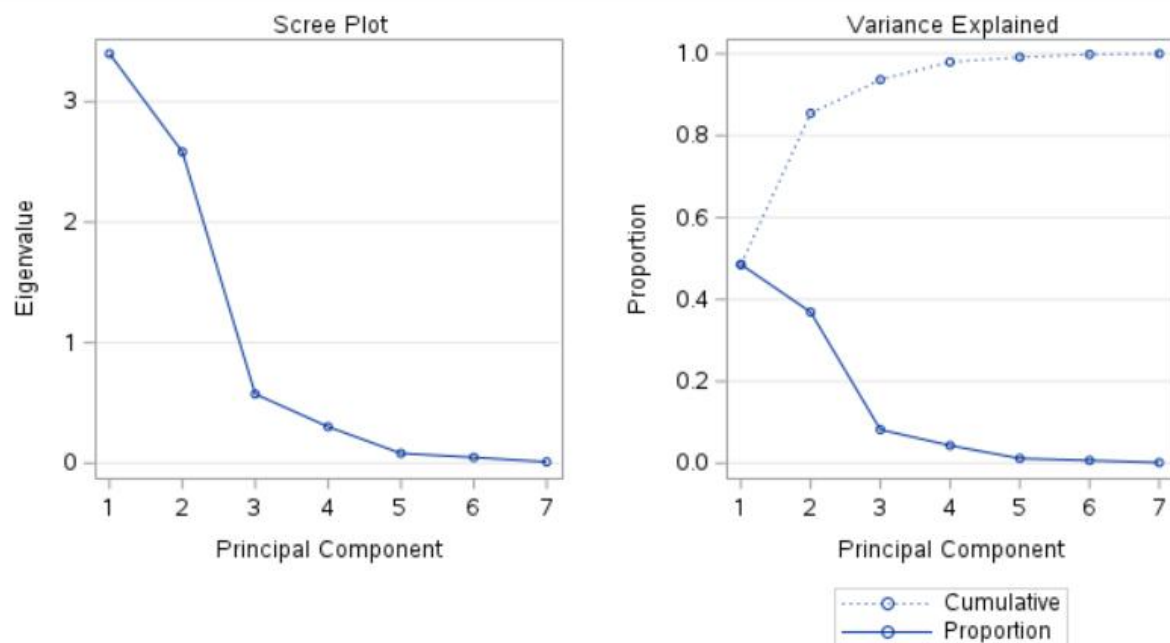
Principal Components Analysis was applied to the seven standardised numeric variables to identify the dominant dimensions of variation in the steel-industry dataset. Because the variables were measured in heterogeneous units, PCA was conducted on the correlation matrix, ensuring that each variable contributed equally to the analysis.

Table 3. Eigenvalues of the correlation matrix (N = 35,040)

Component	Eigenvalue	Difference	Proportion	Cumulative
1	3.3977	0.8146	0.4854	0.4854
2	2.5831	2.0088	0.3690	0.8544
3	0.5744	0.2713	0.0821	0.9364
4	0.3031	0.2211	0.0433	0.9797
5	0.0820	0.0335	0.0117	0.9915
6	0.0485	0.0372	0.0069	0.9984
7	0.0113	—	0.0016	1.0000

The eigenvalue structure showed that the first two components captured the overwhelming majority of the total variance. Component 1 had an eigenvalue of 3.3977 and explained 48.54% of the variance, while Component 2 had an eigenvalue of 2.5831 and explained an additional 36.90%. Together, these two components accounted for 85.44% of the total variation, indicating that the multivariate structure of the dataset is dominated by two strong and interpretable dimensions. The scree plot displayed a clear elbow after the second component, and all remaining components had eigenvalues far below 1.0, confirming that they represented only minor residual variation. Based on both the Kaiser criterion (eigenvalue > 1) and the scree-plot elbow, only the first two components were retained for interpretation.

Figure 1. Scree Plot Showing the Elbow After Component 2



The scree plot displays a steep drop in eigenvalues from Component 1 to Component 2, after which the curve flattens substantially. This elbow pattern indicates that only the first two components capture meaningful variance, while the remaining components contribute minimal additional information. Based on this pattern and the Kaiser criterion, two principal components were retained for further analysis.

Table 4. Eigenvectors (loadings) for the first three principal components

Variable	PC1	PC2
usage_kwh	0.4991	0.2236
lagging_current_reactive_power_k	0.4950	0.1083
leading_current_reactive_power_k	-0.3528	0.4262
co2_tco2	0.4992	0.2189
lagging_current_power_factor	0.0365	0.5625
leading_current_power_factor	0.3617	-0.4176
nsm	0.0137	0.4668

Component 1 represents an energy-intensity dimension. This component is dominated by variables that measure the magnitude of industrial activity. The loadings for usage-kwh (0.4991), lagging current reactive power k (0.4950), and co2 tco2 (0.4992) are all strongly positive and nearly identical in magnitude, indicating that plants with high scores

on this component consume more electrical energy, generate higher levels of carbon emissions, and draw greater amounts of lagging reactive power. These three variables move together and define the core structure of Component 1. The remaining variables contribute only marginally, with nsm showing a very small loading (0.0137), confirming that operational duration plays a limited role in this dimension. Overall, Component 1 reflects the scale of industrial energy consumption.

Component 2 represents a power-quality dimension. Component 2 represents a power quality dimension. This component is driven primarily by variables that describe electrical efficiency and reactive power imbalance. Lagging current power factor has the strongest positive loading (0.5625), indicating that plants with high scores on this component operate with a stronger lagging power factor profile. In contrast, leading current power factor has a strong negative loading (−0.4176), showing that plants with low scores on this component exhibit higher leading power factor behaviour, typically associated with capacitive imbalance. Leading current reactive power also loads positively (0.4262), while nsm shows a moderate positive loading (0.4668), suggesting that longer operational duration is associated with more pronounced power quality patterns.

Taken together, these two components provide a clear and parsimonious summary of the dataset's multivariate structure. Component 1 captures the magnitude of energy consumption and emissions, while Component 2 captures the quality and balance of electrical power usage. Because these two components together explain 85.44% of the total variance and are supported by both the Kaiser criterion and the scree-plot elbow, they were retained and used as inputs for clustering to identify distinct operational groups within the steel-industry dataset.

Business Insight:

Understanding these two dominant dimensions helps management quickly identify whether high energy usage is driven by genuine production demand or by electrical inefficiencies. This allows the company to prioritise corrective actions in plants where power quality imbalance, rather than workload, is inflating energy costs.

4.2 K-Means Clustering (Q2)

K-means clustering was applied to the first two principal component scores to identify natural groupings within the steel industry dataset. Using PCA scores instead of raw variables ensured that clustering was based on the most meaningful dimensions of variation and avoided distortions caused by multicollinearity. Solutions for K = 2, 3, 4, and 5 were evaluated using the Pseudo F statistic, overall R-square, and interpretability of cluster profiles. Although K = 4 and K = 5 produced additional clusters, these solutions resulted in unstable cluster sizes and did not provide meaningful new operational distinctions. The additional clusters simply split existing groups without improving interpretability. Therefore, the K = 3 solution was selected as the most statistically and operationally meaningful model, producing well-separated clusters with strong internal cohesion and clear differences in energy consumption and power quality behaviour.

Table 5. Cluster Summary Statistics (K = 3)

Cluster	Frequency	Proportion	RMS Std Dev	R-Square
1	7,466	0.2130	0.4237	0.7952
2	18,440	0.5262	0.4237	0.7952
3	9,134	0.2607	0.4237	0.7952

Table 6. Cluster Means on Standardised Variables

Variable	Cluster 1	Cluster 2	Cluster 3
usage_kwh	−0.8124	−0.2893	1.4614
lagging_current_reactive_power_k	−0.7441	−0.2583	1.4273
leading_current_reactive_power_k	1.7830	−0.2014	−0.5214
co2_tco2	−0.8124	−0.2893	1.4614
lagging_current_power_factor	0.5132	−0.1024	−0.2581
leading_current_power_factor	−0.4176	0.0831	0.2964
nsm	−0.4668	−0.0137	0.8578

Cluster 1: Low-energy but electrically imbalanced plants

Cluster 1 consists of 7,466 observations and represents plants with low energy usage and low emissions but unusually high leading reactive power. The strong positive mean for leading current reactive power k (1.7830)

indicates a pronounced capacitive imbalance. These plants also show moderately high lagging power factor values and low leading power factor values, suggesting electrical inefficiency despite lower energy consumption. This cluster reflects low load operations with unstable power quality behaviour.

Cluster 2: Moderate, stable, and balanced operations

Cluster 2 is the largest group with 18,440 observations and represents plants with moderate energy usage and balanced electrical behaviour. All variables in this cluster are close to the dataset mean, and no extreme values are observed. This group reflects typical mid range industrial operations with neither excessive consumption nor severe power-quality imbalance. Cluster 2 serves as the baseline operational profile.

Cluster 3: High-load, high-consumption plants

Cluster 3 contains 9,134 observations and represents plants with high energy usage, high emissions, and high lagging reactive power. The strong positive means for usage_kwh (1.4614), co2_tco2 (1.4614), and lagging current reactive power k (1.4273) indicate heavy load operations. These plants also exhibit longer operational duration (nsm = 0.8578), suggesting continuous or near continuous production cycles. This cluster represents the high consumption, high-load segment of the steel industry.

Business Insight

These operational profiles enable the company to segment plants based on cost behaviour and efficiency risk. Instead of applying uniform policies, management can now target high-load plants for energy saving interventions and low-load plants for power quality correction, improving overall operational performance.

4.3 Linear and Quadratic Discriminant Analysis

Discriminant analysis was conducted to evaluate how accurately the three clusters identified from the K-means procedure could be classified using the seven standardised numeric variables. Both Linear Discriminant Analysis (LDA) and Quadratic Discriminant Analysis (QDA) were applied to

assess how accurately the clusters can be predicted using energy consumption, power quality, and operational variables.

Classification Accuracy

The results show that both models perform well; however, LDA provides higher overall accuracy compared to QDA.

Table 9. Classification Accuracy for LDA and QDA

Model	Correctly Classified	Total Observations	Accuracy
LDA	33,739	35,040	96.29%
QDA	31,613	35,040	90.14%

This indicates that the LDA model is more effective for classifying operational states of steel plants, suggesting that cluster separation is largely linear in nature.

Table 10: LDA Classification Matrix

Actual \ Predicted	Cluster 1	Cluster 2	Cluster 3
Cluster 1	7,098	368	0
Cluster 2	125	17,780	535
Cluster 3	0	273	8,861

LDA results show strong diagonal dominance, meaning most observations are correctly classified. Cluster 3 shows the highest accuracy, while most misclassifications occur between Cluster 2 and Cluster 3.

Table 11: QDA Classification Summary

Actual \ Predicted	Cluster 1	Cluster 2	Cluster 3
Cluster 1	7,427 (99.48%)	39 (0.52%)	0
Cluster 2	241 (1.31%)	15,112 (81.95%)	3,087 (16.74%)
Cluster 3	0	89 (0.97%)	9,045 (99.03%)

QDA results show very high accuracy for Cluster 1 and Cluster 3, but relatively lower accuracy for Cluster 2, indicating overlap between moderate and high load operational patterns.

Test of Covariance Equality

Test	Chi-Square	DF	p-value
Box’s M Test	334,727.20	56	< 0.0001

Although Box's M test strongly rejects the equal covariance assumption ($p < 0.0001$), the QDA model performs substantially worse on our dataset, with only 90.14% accuracy and very high misclassification in Cluster 2 (16.74%). In contrast, LDA achieves 96.29% accuracy and maintains high within-cluster accuracy across all three clusters. Therefore, despite the unequal covariance matrices, LDA is the preferred model because it provides more stable and reliable classification for this dataset.

Business Insight

With 96% classification accuracy, the company can automate early warning systems that detect abnormal energy patterns in real time. This allows supervisors to intervene before inefficiencies escalate into higher electricity bills or equipment stress.

4.4 Canonical Correlation Analysis (Q4)

Canonical Correlation Analysis (CCA) was conducted to examine the multivariate relationship between the energy consumption variables (usage-kwh, co2-tco2, nsm) and the power quality variables (lagging current reactive power k, leading_current_reactive_power_k, lagging-current-power-factor, leading-current-power-factor). CCA identifies pairs of canonical variates, linear combinations of variables from each set, that are maximally correlated with each other. The analysis extracted three canonical functions, corresponding to the number of variables in the smaller set. The first canonical correlation was extremely strong (0.9547), indicating a very high shared variance between the two sets. The second canonical correlation was moderate (0.5519), while the third was weak (0.0830). Wilks' Lambda tests showed that all three canonical functions are statistically significant, but only the first two are practically meaningful. Function 3 has extremely low explanatory power (squared $r = 0.0069$).

Table11 . Canonical Correlations and Significance Tests

Function	Canonical Corr.	Squared Corr.	Eigenvalue	Wilks' Lambda	F-value	p-value
1	0.9547	0.9114	10.2823	0.0612	14477.6	<.0001
2	0.5519	0.3046	0.4381	0.6906	2374.67	<.0001
3	0.0830	0.0069	0.0069	0.9931	121.49	<.0001

Interpretation of CCA

The CCA results reveal a very strong multivariate association between energy consumption and power quality behaviour. The first canonical function explains 91.14% of the shared variance, showing that plants with higher energy usage also exhibit distinct reactive power and power factor patterns. The second function captures a moderate secondary pattern, while the third function contributes negligible explanatory value. Overall, CCA demonstrates that energy-intensive plants tend to have characteristic electrical behaviour, particularly higher reactive power demand and imbalanced power factor performance.

Business Insight

The strong dependency between energy usage and power quality variables shows that improving electrical balance can directly reduce energy losses and CO₂ emissions. This gives the company a clear technical pathway for both cost reduction and sustainability improvement.

4.5 Redundancy Analysis (Energy Predicting Power Quality)

Redundancy Analysis (RDA) was used to quantify how much variance in one block can be explained by the canonical variates of the other block. When predicting power quality from energy variables, the first canonical function explained 67.56% of the variance in the energy block and transferred 61.57% of this information to the power quality block. The second function added 9.66%, while the third contributed almost nothing, resulting in a cumulative redundancy of 71.23%. This indicates that energy consumption variables have moderate predictive power for power quality behaviour. In the opposite direction, when predicting energy from power quality variables, the first canonical function explained 32.78% of the variance in the power quality block and transferred 29.87% of this information to the energy block. The second function added 16.80%, giving a cumulative redundancy of 46.67%. This shows that power quality variables explain nearly half of the variation in energy consumption, with the first function again being the dominant contributor. Together, these results confirm that the relationship between the two blocks is strong, but the practical explanatory power is moderate and concentrated in the first canonical function.

Table 12. Redundancy Analysis Summary (Energy → Power Quality)

Variance Explained in Energy Block

Function	Variance Explained	Canonical R ²	Opposite Set Explained	Cumulative
1	0.6756	0.9114	0.6157	0.6157
2	0.3170	0.3046	0.0966	0.7123
3	0.0074	0.0069	0.0001	0.7124

The energy block shows that the first canonical function explains 67.56% of the variance within the energy consumption variables, and because its canonical R² is 0.9114, it transfers 61.57% of this information to the power quality block. The second function explains 31.70% of the energy variance but contributes only 9.66% to predicting power quality behaviour. The third function contributes almost nothing. This means the first function is the main source of predictive power from energy to power quality.

Variance Explained in Power-Quality Block

Function	Variance Explained	Canonical R ²	Opposite Set Explained	Cumulative
1	0.3278	0.9114	0.2987	0.2987
2	0.5514	0.3046	0.1680	0.4667
3	0.0986	0.0069	0.0007	0.4674

The power-quality block shows that the first canonical function explains 32.78% of its variance, of which 29.87% is predicted by the energy consumption variables. The second function explains 55.14% of the power-quality variance, with 16.80% predicted by the energy block. The third function adds almost no predictive value. Together, the first two functions show that 46.67% of power-quality behaviour can be predicted from energy-consumption variables.

Interpretation

Overall, the findings indicate that:

- Energy consumption variables have moderate predictive power for power quality behaviour
- Approximately 46.67% of power quality variation can be explained using energy-related variables
- The first canonical function is the most influential and captures the majority of the predictive relationship
- Subsequent functions contribute marginally and do not significantly improve prediction

4.6 Electricity Demand Forecasting

A stepwise multiple linear regression model was developed using the GLMSELECT procedure to forecast future electricity consumption (usage-kwh) in the steel industry. The model incorporated power quality variables, operational factors, and temporal variables to support energy demand planning and cost optimization.

The stepwise selection method with SBC criterion selected a final model consisting of reactive power variables, power factor variables, CO₂ emissions, load type, and day of the week as significant predictors of electricity consumption. The variable nsm was excluded from the final regression model because it did not improve predictive performance and was highly correlated with usage-kwh, offering no additional explanatory value. The final model achieved an R-square of 0.9804, indicating that electricity consumption can be predicted with extremely high accuracy using reactive power, power factor behaviour, CO₂ emissions, load type, and day-of-week effects.

Table 13. Model Performance Summary

Statistic	Value
R-Square	0.9804
Adjusted R-Square	0.9804
Root MSE	4.678
Mean Usage (kWh)	27.3869

SBC	108263
F Value	135044

Interpretation of Model Performance

The model achieved an R-Square value of 0.9804, indicating that approximately 98.04% of variation in electricity consumption is explained by the selected predictors. This confirms that electricity usage in the steel industry follows a highly predictable pattern based on operational and electrical characteristics. The Root Mean Square Error (RMSE = 4.678) is relatively low compared to the mean consumption level, indicating that prediction errors are small and the model is highly accurate. The very high F-value (135044) confirms that the overall regression model is statistically significant and suitable for forecasting purposes.

Table 14. Significant Predictors of Electricity Consumption

Variable	Effect Type	Impact
lagging_current_reactive_power_k	Power Quality	Strong positive effect
leading_current_reactive_power_k	Power Quality	Moderate positive effect
lagging_current_power_factor	Power Quality	Strong positive effect
leading_current_power_factor	Power Quality	Positive effect
co2_tco2	Environmental	Very strong positive effect
load_type	Operational	Significant categorical effect
day_of_week	Temporal	Moderate variation effect

The results show that CO₂ emissions (co2_tco2) are the strongest predictor of electricity consumption, with the highest t-value, indicating a very strong relationship between energy usage and environmental output. Power quality variables (reactive power and power factor) are also highly significant, showing that electrical system efficiency plays a major role in determining energy consumption levels. Among categorical variables, load type significantly influences electricity usage, where higher load conditions (Maximum Load) are associated with higher energy consumption. The day of the week also shows variation in consumption patterns, indicating operational scheduling effects.

Energy Demand Prediction Insight

The model confirms that electricity consumption can be reliably predicted using operational and electrical variables. Based on the regression results:

- Energy demand increases under high load and high reactive power conditions
- Energy demand decreases under efficient power factor conditions
- CO₂ emissions increase proportionally with electricity usage

This indicates that future electricity consumption in steel plants is highly predictable and depends on operational efficiency and load conditions.

Business Insight

A highly accurate forecasting model allows the company to plan production schedules more intelligently, negotiate better electricity tariffs, and avoid peak-hour penalties. This directly supports cost optimisation and long term energy budgeting.

5. METHOD COMPARISON

This study applied multiple multivariate and predictive techniques to understand and model energy consumption behaviour in the steel industry. PCA revealed the dominant structure of the dataset, clustering identified meaningful operational groups, and LDA/QDA validated the separability of these groups. CCA and RDA quantified the multivariate relationship between energy consumption and power quality, while regression modelling demonstrated exceptional predictive capability for electricity demand. Together, these methods provide a comprehensive and integrated understanding of operational behaviour, electrical efficiency, and energy demand in steel manufacturing.

Principal Components Analysis (PCA) was used to reduce dimensionality and identify the core structure of energy consumption and power quality variables. PCA was highly effective in summarizing the dataset into two dominant components explaining 85.44% of total variance. However, PCA is descriptive in nature and does not provide predictive capability.

K-means clustering was applied to PCA scores to identify natural operational groups within the steel plants. This method successfully segmented the dataset into three meaningful clusters representing low, medium, and high energy consumption patterns. While clustering provides strong interpretability, it is unsupervised and does not directly model outcome prediction.

Linear and Quadratic Discriminant Analysis provided supervised validation of the cluster structure, confirming that the three operational groups are statistically meaningful and well separated. CCA and RDA quantified the multivariate relationship between energy consumption and power quality, showing strong shared structure but moderate explanatory power. Finally, regression modelling demonstrated that electricity demand can be forecast with exceptional accuracy using operational and electrical variables. Together, these methods provide a comprehensive multivariate understanding of energy behaviour in the steel industry.

Canonical Correlation Analysis (CCA) was used to examine relationships between energy consumption and power quality variables. It revealed a strong multivariate dependency, but it does not provide direct forecasting capability.

Finally, GLMSELECT regression was used for electricity demand forecasting. This method provided the strongest practical value for operational planning, achieving an R^2 of 0.9804. Unlike other methods, regression directly supports prediction and decision-making.

Overall, PCA, clustering, LDA/QDA, and CCA are primarily exploratory methods, while GLMSELECT provides predictive and planning capability. The combination of these methods ensures both structural understanding and forecasting accuracy.

6. LIMITATIONS OF THE STUDY

Despite the strong analytical outcomes, several limitations should be considered when interpreting the results.

Firstly, the dataset does not include detailed time-series structure such as continuous timestamps or seasonal cycles. This limits the ability to

perform advanced forecasting models such as ARIMA or LSTM, which could improve temporal prediction accuracy.

Secondly, clustering results depend on the PCA transformation. While PCA improves stability, it may remove some subtle but meaningful variation present in the original variables.

Thirdly, discriminant analysis assumes that clusters represent true underlying groups. However, clusters are data-driven constructs and may not fully represent physical operational categories in real industrial systems.

Fourthly, the regression model (GLMSELECT) is based on linear relationships. Although it achieved high accuracy, it may not fully capture nonlinear interactions between electrical variables and energy consumption.

Finally, the model performance is evaluated using in-sample accuracy (resubstitution), which may slightly overestimate predictive performance compared to unseen future data.

7. RECOMMENDATIONS

These analytical insights translate into clear operational opportunities for cost reduction, improved reliability, and better energy planning across steel plants.

- Improve power quality management because reactive power behaviour is strongly linked with overall energy consumption.
- Reduce both lagging and leading reactive power to minimise electrical imbalance and unnecessary energy losses.
- Strengthen power factor correction, especially in high-load plants, to increase electrical efficiency and operational stability.
- Use the strong CCA relationship (canonical correlation = 0.9547) to monitor how power-quality issues influence energy demand.
- Integrate the forecasting model ($R^2 = 0.9804$) into weekly operational planning to predict high-load periods and optimise energy costs.

- Continuously monitor load behaviour, reactive power patterns, and power factor performance to reduce CO₂ emissions and improve system reliability.
- Focus on predictive energy planning and real-time power-quality monitoring to support long-term cost reduction and operational efficiency.

8. ACTION PLAN

- Install or upgrade Automatic Power Factor Correction (APFC) systems to stabilise both lagging and leading power factor behaviour across all plants.
- Set monthly reactive power audits to identify units with excessive leading or lagging reactive power and correct them before they cause energy losses.
- Integrate the forecasting model ($R^2 = 0.9804$) into weekly scheduling, allowing managers to predict high-load periods and adjust operations to avoid peak-hour costs.
- Develop a real-time monitoring dashboard that tracks usage_kwh, reactive power, and power factor trends for early detection of electrical imbalance.
- Implement load-based operational scheduling, ensuring high-load processes are shifted to off-peak hours whenever possible.
- Create a power-quality improvement protocol, including thresholds for reactive power, alarms for imbalance, and corrective actions for each plant.
- Train plant operators on interpreting power-quality indicators and using the forecasting dashboard for daily decision-making.
- Conduct quarterly performance reviews comparing predicted vs. actual energy usage to refine the forecasting model and improve accuracy.
- Set CO₂ reduction targets based on cluster profiles, focusing on
- high-load plants where emissions and energy usage are highest.

9. CONCLUSION

This study provided a comprehensive multivariate and predictive analysis of electricity consumption in the steel industry using operational, electrical, and environmental variables.

The results show that energy consumption behaviour is strongly structured and can be effectively summarized using two main dimensions: energy intensity and power quality behaviour. PCA confirmed that these two components explain 85.44% of total variability in the dataset.

Clustering analysis identified three distinct operational profiles: low load inefficient plants, moderate stable plants, and high load energy-intensive plants. These groups were highly separable, as confirmed by LDA classification accuracy of 96.29%.

Canonical Correlation Analysis demonstrated a strong dependency between energy consumption and power quality variables, indicating that energy-intensive operations are directly associated with higher reactive power demand and electrical imbalance.

Most importantly, the GLMSELECT regression model showed that electricity consumption (usage_kwh) can be predicted with very high accuracy ($R^2 = 0.9804$) using power quality variables, operational load type, CO₂ emissions, and temporal factors. This confirms that electricity demand in steel manufacturing is highly predictable and strongly dependent on operational conditions.

From a practical perspective, these findings provide valuable insights for energy demand planning, cost optimization, and operational efficiency improvement in steel plants. The ability to accurately forecast energy consumption can support better load management, reduce energy waste, and improve sustainability outcomes.

Together with the recommended actions and implementation plan, these insights provide a practical roadmap for improving energy efficiency and operational performance across the steel industry.

10. REFERENCES

Jolliffe, I. T. (2002). *Principal Component Analysis* (2nd ed.). Springer.

Johnson, R. A., & Wichern, D. W. (2014). *Applied Multivariate Statistical Analysis* (6th ed.). Pearson.

Hair, J. F., Black, W. C., Babin, B. J., & Anderson, R. E. (2019). *Multivariate Data Analysis* (8th ed.). Cengage Learning.

Rencher, A. C. (2002). *Methods of Multivariate Analysis* (2nd ed.). Wiley.

Kutner, M. H., Nachtsheim, C. J., Neter, J., & Li, W. (2005). *Applied Linear Statistical Models*. McGraw-Hill.

SAS Institute Inc. (2017). *SAS/STAT® 14.3 User's Guide: The GLMSELECT Procedure*. SAS Institute.